

Does an AGN's infrared spectrum remember its accretion history?

Scientific Category: Supermassive Black Holes and Active Galaxies

Alternate Category: Interstellar Gas and Dust

Scientific Keywords: AGN host galaxies, Interstellar dust, Quasars

Instruments: MIRI

Proposal Size: Medium

Exclusive Access Period: 12 months

Allocation Information (in hours):

Science Time: 17.1

Charged Time: 67.0

Abstract

An AGN can fade while its warm dust still glows from an earlier bright state, and after it brightens the dust can lag behind the new nuclear power. How long does this infrared memory last, and is dust replenished after a nucleus fades? We request 67.0 hours of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$ with recorded rises, declines, flares and recoveries. ZTF and decade-long WISE monitoring, SPHEREx measurements and multi-epoch optical spectroscopy tie each nuclear history to its hot dust. MIRI supplies what no other facility now can: the 5-28 micron warm continuum, both silicate features, and the PAH and spatial diagnostics that separate the nucleus from its host. Anchored by 2010 WISE 12 and 22 micron photometry, we will jointly fit each spectrum with its activity history and test whether nuclei at comparable present power carry different warm spectra because they reached that power by different paths. Delayed heating and cooling of a fixed dust distribution is the null hypothesis. Fading and recovering sources test whether dust must be destroyed, replenished or redistributed; rising sources provide the comparison. Together the 24 spectra constrain the response timescales and the radial extent of the responding dust. The outcome is a bound on how fast the dusty structure around an AGN can change, and a measure of how long the mid-infrared emission of a changed nucleus misrepresents its present state, which matters for surveys that use mid-infrared light to gauge AGN power and obscuration.

Target Summary:

Target	RA	Dec
F01	07 59 34.9500	+32 21 43.30
F01-SKY	07 59 39.6840	+32 19 59.37
F02	00 33 52.7100	+20 21 4.10
F02-SKY	00 33 56.8310	+20 21 19.63
F03	10 53 25.4100	+30 24 19.30
F03-SKY	10 53 33.4960	+30 24 28.44
F04	00 53 46.1600	+22 32 22.40
F04-SKY	00 53 43.9190	+22 30 26.49
F05	08 53 59.0600	+21 28 3.40
F05-SKY	08 54 1.2850	+21 29 59.31
F06	01 25 58.0600	+15 18 13.70
F06-SKY	01 25 55.1270	+15 18 56.13
F07	23 44 33.0900	+26 00 38.90
F07-SKY	23 44 38.1330	+26 01 10.59
F08	10 44 27.7400	+27 18 6.50
F08-SKY	10 44 32.0880	+27 18 22.02
F09	08 35 38.7800	-04 05 17.50
F09-SKY	08 35 34.7700	-04 05 17.50
F10	08 53 37.2600	+01 43 3.60
F10-SKY	08 53 30.9130	+01 42 19.22
F11	23 30 32.3000	-02 27 45.00
F11-SKY	23 30 29.2970	-02 26 27.06
F12	04 58 40.3000	-21 59 31.00
F12-SKY	04 58 35.4470	-22 00 51.43
R01	00 46 7.9766	+09 07 20.95
R01-SKY	00 46 5.2050	+09 09 13.71
R02	22 05 37.7100	-07 11 14.50
R02-SKY	22 05 29.9220	-07 10 43.44
R03	10 47 5.1600	+54 44 5.90
R03-SKY	10 46 59.9610	+54 45 23.84
R04	08 29 42.6700	+41 54 36.90
R04-SKY	08 29 33.3630	+41 53 36.88
R05	10 40 43.3900	+27 35 19.00
R05-SKY	10 40 40.1990	+27 34 36.57
R06	08 38 26.5000	+37 19 6.70
R06-SKY	08 38 22.6460	+37 19 45.26

Does an AGN's infrared spectrum remember its accretion history?

Target	RA	Dec
R07	10 11 52.9800	+54 42 6.40
R07-SKY	10 11 52.9800	+54 43 6.40
R08	09 48 38.4200	+40 30 43.60
R08-SKY	09 48 42.4500	+40 30 5.03
R09	06 55 49.5300	+40 00 1.20
R09-SKY	06 55 42.7480	+39 59 16.19
R10	19 26 30.2200	+41 33 5.00
R10-SKY	19 26 34.0530	+41 34 6.43
R11	19 49 13.7000	-18 47 16.00
R11-SKY	19 49 17.9090	-18 47 21.23
R12	20 14 59.2600	+25 23 1.20
R12-SKY	20 15 0.1280	+25 20 46.71

Observing Summary:

Target	Observing Template	Flags	Allocation *
F01	MIRI Medium Resolution Spectroscopy		1,268 / 6,492
F01-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
F02	MIRI Medium Resolution Spectroscopy		1,268 / 5,064
F02-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F03	MIRI Medium Resolution Spectroscopy		1,268 / 4,969
F03-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,870
F04	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
F04-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
F05	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
F05-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
F06	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
F06-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F07	MIRI Medium Resolution Spectroscopy		1,268 / 6,609

Does an AGN's infrared spectrum remember its accretion history?

Target	Observing Template	Flags	Allocation *
F07-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,838
F08	MIRI Medium Resolution Spectroscopy		1,268 / 4,843
F08-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F09	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
F09-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F10	MIRI Medium Resolution Spectroscopy		1,268 / 5,229
F10-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,870
F11	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
F11-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,855
F12	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
F12-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,870
R01	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R01-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
R02	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R02-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
R03	MIRI Medium Resolution Spectroscopy		1,268 / 6,492
R03-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,855
R04	MIRI Medium Resolution Spectroscopy		1,268 / 4,949
R04-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
R05	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R05-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R06	MIRI Medium Resolution Spectroscopy		1,492 / 6,752
R06-SKY	MIRI Medium Resolution Spectroscopy		1,492 / 4,079
R07	MIRI Medium Resolution Spectroscopy		1,268 / 5,062

Does an AGN's infrared spectrum remember its accretion history?

Target	Observing Template	Flags	Allocation *
R07-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R08	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R08-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R09	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R09-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,855
R10	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R10-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,838
R11	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R11-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R12	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R12-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,896

* Science duration / charged duration (sec)

Total Prime Science Time in Hours: 17.1

Total Charged Time in Hours: 67.0

Observation Description, Investigators, and Team Expertise are all included in this preview for your team to review. These will not appear in the version of the proposal given to the TAC. The Observation Description is only needed by STScI operations, and the investigator information is removed to allow for a dual anonymous review.

Observing Description

We request 67.0 hours of MIRI Medium Resolution Spectroscopy (MRS) for 24 nearby ($z < 0.3$) changing-look and strongly variable AGN with recorded rises, declines, flares and recoveries. Each nucleus will be observed once in all three MRS grating settings, providing approximately 4.9–27.9 micron coverage of the warm continuum, both silicate features, PAH bands and diagnostic emission lines. Science observations use nuclear target acquisition, four-point dithers and FULL/FASTR1 readout. The requested continuum sensitivity is $S/N = 50$ per $R = 100$ bin near rest-frame 5 and 12 microns and $S/N = 30$ near 18 microns. The faintest W3 target, J083826.50+371906.7, receives additional exposure in setting B.

Each science visit has an equal-depth observation of a nearby blank-sky field selected using WISE and 2MASS data. Separate sky observations preserve extended host emission for spatial and spectral decomposition. Sky visits use four-point extended-source dithers without target acquisition. Each science/sky pair is linked as a consecutive, noninterruptible sequence to provide a contemporaneous background. The programme comprises 24 science and 24 sky visits, with 17.00 hours of science-plus-sky integration and 67.0 hours total charged time. No fixed-date constraints, coordinated parallels or joint-facility time are requested.

Investigators:

Role	Investigator	Institution	Country
CoI	Dr. Andreas L Faisst	California Institute of Technology	USA/CA
CoI	Dr. George Helou	California Institute of Technology	USA/CA
PI &	Dr. Shoubaneh Hemmati	California Institute of Technology	USA/CA
CoI	Dr. Jessica Krick	California Institute of Technology	USA/CA
CoI	Dr. Yu-Heng Lin	California Institute of Technology	USA/CA
CoI *	Dr. Irene Shivaiei	Centro de Astrobiologia - CAB	ESP
CoI	Dr. David L Shupe	Caltech/IPAC	USA/CA

Number of investigators: 7

* ESA investigators: 1

& Contacts: 1

Team Expertise:

Our team combines AGN variability, infrared dust physics, MIRI/MRS spectroscopy and the calibration of the surveys that supply the activity histories. The ancillary data are in hand: ZTF and WISE/NEOWISE light curves for all 24 targets, SPHEREx spectra already extracted for the sample, archival multi-epoch optical spectra, and Palomar/NGPS spectra obtained in 2026. Most of the team is based at Caltech/IPAC, which hosts the IRSA archives for WISE, ZTF and SPHEREx.

PI Shoubaneh Hemmati developed the machine-learning light-curve analysis used to select the sample (Hemmati et al. 2026, ApJ, 998, 130), leads the Palomar/NGPS follow-up, and will coordinate reconstruction of the activity histories and the joint tests of dust memory and evolution.

Does an AGN's infrared spectrum remember its accretion history?

Irene Shivaiei served on the JWST/MIRI development and commissioning teams and leads MIRI/MRS analysis of PAH emission in the JWST program PAHSPECS; she will guide MRS reduction, spectral decomposition and interpretation of resolved host dust.

George Helou brings decades of work on infrared emission, dust heating and the separation of star-formation and AGN contributions, and will lead the physical interpretation of the warm continuum and aromatic features together with the dust-response modelling.

Jessica Krick brings Spitzer/IRAC calibration expertise, including infrared photometry of JWST calibration stars, and will lead the synthetic-photometry comparison with the 2010 WISE baseline and the assessment of background and calibration uncertainties.

Andreas Faisst, a SPHEREx science-team member with dust-modelling experience and a leading role in cosmos-web, will validate the SPHEREx spectra and support joint modelling of nuclear and host emission.

David Shupe brings ZTF pipeline experience in astrometric calibration, source matching and relative photometry, and will handle light-curve quality control and reproducible analysis.

Yu-Heng Lin brings quasar spectroscopy and emission-line diagnostics, and will measure the optical spectral states for comparison with the infrared diagnostics.

■ Scientific Justification (required for all)

Does the dust around a changing nucleus keep up with it? An active galactic nucleus (AGN) can undergo large luminosity changes within years [1,2], sometimes accompanied by a transformation of its continuum and broad-line spectrum [3–5]. Its surrounding dust records these changes through delayed heating and through destruction or replenishment of the emitting material. Individual nuclei show that dust boundaries evolve over years [6,7]; the missing step is a sample that reveals how the response depends on the direction and history of nuclear activity. Changing-look and strongly variable AGN followed through rises, declines, flares and recoveries provide exactly this experiment.

We request **67.0 h of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$** with recorded rises, declines, flares and recoveries. ZTF/WISE monitoring, SPHEREx coverage and archival plus 2026 optical spectroscopy connect nuclear activity to the hot-dust response. Together they catch dust at different stages of its response. Only MIRI now covers the warm continuum and both silicate features, with the spatial information that separates nucleus from host, tying each history to its warm-dust spectrum. The programme addresses two linked questions:

1. How long does warm-dust memory persist? Delayed heating can produce a warm excess after a decline and a deficit after a rise [8,9]. The recorded histories and 2010 WISE measurements constrain response models and test whether this spectral memory persists at the MIRI epoch.

2. Does dust replenish after the nucleus fades? Delayed heating and cooling of unchanged dust is the null model. The silicate features and 5–28 μm continuum test whether fading and recovering sources require an altered dust distribution [6–9], with rising sources providing the comparison.

From infrared echoes to dust evolution. WISE observations have established infrared echoes of changing-look transitions [10–13], supporting intrinsic changes in nuclear power in those sources. Broadband colours, however, combine temperature, emitting area, extinction and host light [14–16]. The proposed spectra separate these contributions, measuring the warm signature of each history and constraining which properties of the emitting material produce it.

The equilibrium sublimation radius scales approximately as $L^{1/2}$ [17–19], but the physical dust boundary can adjust over years [6]. Individual nuclei already suggest different responses: the smaller dust radius after Mrk 590 faded supports replenishment [7], whereas the temperature variations in Mrk 817 favour heating and cooling at a nearly fixed boundary [20]. Fading and recovering sources therefore target the uncertain replenishment of dust, with rising sources testing the response to increased illumination. The comparison asks whether the spectrum depends on the path to its present state.

MIRI’s diagnostic power. SPHEREx covers 0.75–5 μm [21], constraining hot dust close to the nucleus [20,22]. Longer delays and weaker variability at longer wavelengths [8,23,24] make the warm continuum sensitive to earlier illumination. MRS supplies the essential 4.9–27.9 μm spectrum [25]: the continuum measures the imprint, the paired silicate features constrain geometry and optical depth [26,27], and spatial profiles and aromatic bands constrain host emission [16,28,29]. MIRI extends the NEOWISE/SPHEREx record into the warm-dust regime and connects it to the 2010 W3/W4 baseline. This combination tests how dust at different temperatures responds to the same recorded activity.

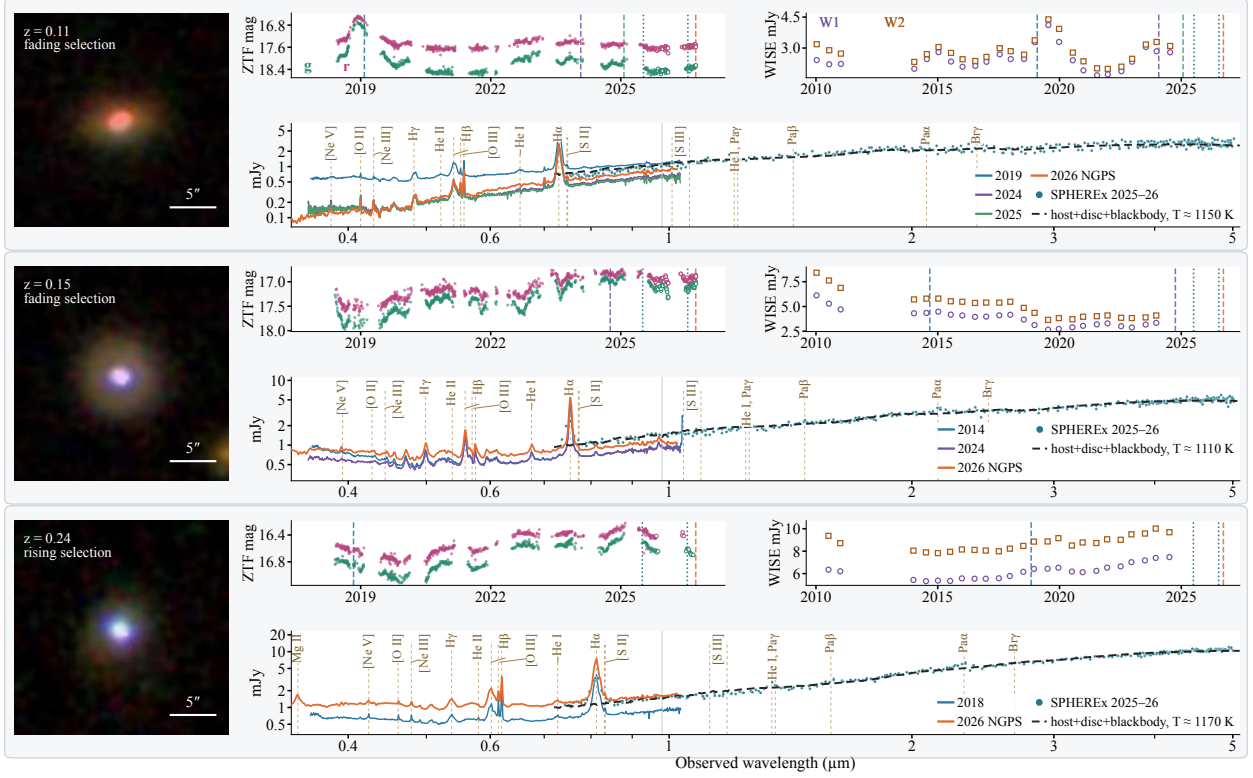


Figure 1: **Three examples of the rich histories assembled for the 24-target sample.** Each example combines ZTF and WISE light curves, archival and 2026 optical spectra and multiple SPHEREx visits, connecting nuclear activity to hot dust. **Top:** a 2018–2019 flare followed by fading; **middle:** an infrared decline with modest optical recovery; **bottom:** rising light curves and optical spectra. MIRI will test how much of this history the warm dust retains. SDSS images have 5'' bars. Open ZTF points are alerts through 2026 September; coloured ticks mark spectral epochs. Dashed curves show mean host+disc+blackbody fits to SPHEREx ($T_{\text{col}} \approx 1100\text{--}1200$ K, model-dependent); gold markers indicate emission lines. Wavelengths are observed-frame; spectra are in mJy.

A sample selected to test dust memory. Our targets span sustained rises, declines, flares, plateaus and recoveries, testing the direction, amplitude and timing of the dust response. Recovering sources ask whether nuclei at comparable current power retain different warm spectra because they followed different paths. We combine catalogued changing-look AGN [30–33] with large-amplitude ZTF/WISE variables [5,34], selecting across spectral classifications [31,35]. A variability manifold identifies candidates with similar light-curve shapes [36]; available multi-epoch spectra establish continuum and broad-line changes. Visual inspection shows no obvious mergers or very close projected companions. Each target has approximately a decade of W1/W2 monitoring, with 95th-to-5th-percentile flux ratios of 1.3–2.3 before host subtraction. SPHEREx measurements supplement these histories, with repeated all-sky passes adding near-infrared epochs

before the MIRI observations.

From dust components to their response. Published observations demonstrate the three elements of our experiment: separating hot and warm dust (Figure 2A), constraining central and surrounding emission (B), and interpreting a delayed infrared response using the activity history (C). The rest-frame 8–13 μm dust luminosity includes broad silicate emission or absorption, with PAHs and narrow lines fitted separately. Continuum slopes and paired silicate profiles constrain the temperature distribution, optical depth and geometry, including when an integrated flux is consistent with its historical value.

Connecting spectral shape to the recorded history. We will jointly fit the warm spectrum, current nuclear power and each source’s full light curve, including reversals. Optical decomposition, WISE and SPHEREx constrain illumination and hot dust at their observed epochs [4,20,22]; the optical photometry nearest each JWST visit constrains the current state, with temporal gaps propagated as uncertainty. Figure 2C illustrates why earlier illumination matters: the warm response can be delayed and broadened. Each target contributes one warm spectral snapshot; the spread of event ages across the sample replaces repeat visits. The 2010 W3/W4 measurements provide additional constraints through their bandpasses and a shared host model. Event dates, gaps and the optical-to-heating conversion enter as uncertainties; an old flare without a warm imprint itself bounds the memory timescale.

Turning histories into a test of dust memory. For each target, we will compare its current nuclear power with its earlier illumination, averaged over the delays allowed by the dust-response models. Sources brighter in the past may retain excess warm emission, whereas recently brightened sources may show an incomplete warm response. Flares and recoveries test whether this dependence persists beyond a simple rising/fading classification. We will test whether the recorded histories explain spectral differences that current power alone cannot, propagating uncertainties in host emission, geometry and unobserved activity.

Measurement precision and interpretation. An illustrative shell-model calculation for the rising source in Figure 1 (bottom), using dust opacities from [37], gives conditional 1σ uncertainties near 5% on the extracted 8–13 μm flux and 0.08/0.04 on the 9.7/18 μm contrasts, defined as $S_\lambda = \ln(F_\lambda/F_{\lambda,\text{cont}})$ relative to a local power-law continuum. These estimates assume the requested S/N, correlated calibration errors of 3% in scale and tilt, and a smooth host residual of 10% at rest 12 μm . At this precision a single object does not separate the two families; the discrimination comes from the ensemble, where 24 histories with different event ages and directions must be reproduced by one response model. Figure 2 shows the observed diagnostics on which the joint analysis rests.

Testing whether delayed heating is sufficient. Clumpy and disc-plus-wind geometries [38–40] define the fixed-distribution null: dust changes temperature in response to the reconstructed history [8,9]. An accretion-fed dusty wind [41] motivates an alternative in which the dust supply and covering factor can change. We will test evolving distributions with dust destruction or replenishment [6,7,22], including light-travel delays and composition and grain-size uncertainty [42] in both families; a preference for evolution would not uniquely identify a wind. We summarize the warm residual by $\delta_{\text{warm}} = \log_{10}[L_{8-13\mu\text{m}}^{\text{MRS}}/L_{8-13\mu\text{m}}^{\text{fixed}}(t_{\text{MIRI}})]$, relative to the *delayed fixed-distribution* prediction. Evidence for evolution requires joint continuum and silicate discrepancies

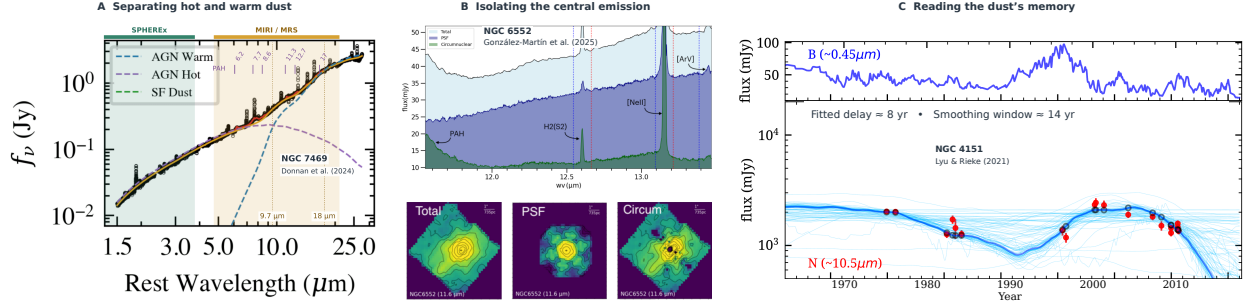


Figure 2: Published examples motivate our experiment across 24 AGN activity histories. A: NGC 7469 illustrates hot/warm dust decomposition [15, Fig. 8]. Green/gold shading marks nominal SPHEREx/MRS rest-wavelength access common to our sample. Black points are data; yellow/red curves are the dust continuum/full model. Purple ticks mark PAH bands; dotted gold lines mark silicate features. **B:** NGC 6552 spectra and 11.6 μm maps demonstrate separation of central and surrounding emission [28, Fig. 4], constraining resolved host light. **C:** NGC 4151 illustrates a delayed, broadened warm-dust response [23, Fig. 15]: blue is optical emission, red points are 10.5 μm measurements, and light-blue curves are fitted responses. This motivates our history-dependent spectral test, which uses event age in place of a measured lag. Wavelengths are rest-frame in A and observed-frame in B. Longer-wavelength data and ancillary fit labels are omitted from C.

across the allowed fixed models [28], robust to host and calibration uncertainties [43], and an improved fit from an evolving distribution. Where both families fit, the data still bound the warm response and the permitted evolution. Across the recorded event ages, these model-dependent constraints test allowed replenishment timescales, the radial extent of responding dust, and how long mid-infrared emission can misrepresent a changed nucleus [44,45].

Complementary tests of obscuration and excitation. Paired silicate profiles, optical histories and high-ionization lines constrain obscuration and excitation [26,27]. [Ne VI] 7.7 and [Ne V] 14.3 μm are accessible throughout the sample [46]; the nominal MRS range includes [Ne V] 24.3 μm for 13 targets and [O IV] 25.9 μm for nine. Line detections or limits provide supporting diagnostics; they are not instantaneous measures of nuclear power.

Central and surrounding dust. Here, surrounding emission denotes the spatially resolved host component after PSF modelling (see Description of Observations). Where detected, we compare it with central emission at matched resolution within the common field. PAH 11.3/7.7 μm ratios, continuum slopes and silicate strengths probe dust processing, heating and obscuration [29,47]. The decomposition in Figure 2B demonstrates how surrounding emission can be isolated to constrain host templates [28], testing whether a central warm excess survives allowed host contributions. This measures the resolved dust environment around the unresolved variable core.

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. Each target receives one MIRI/MRS science visit and a matching sky visit, using all three grating settings, nuclear acquisition and four-point dithers [25,48]. The 4.9–27.9 μm coverage includes both silicate peaks for all 24 targets; the primary rest-frame 8–13 μm diagnostic does not require channel 4C.

Archival total-source W3/W4 fluxes span 1.9–48.4/6.4–82.6 mJy. We target S/N of 50 per $R = 100$ bin near rest 5 and 12 μm and 30 at 18 μm (2%/3.3% statistical errors). APT specifies the FULL/FASTR1 integrations, with extra setting-B exposure for the W3-faint J083826.50+371906.7. Equal-depth sky exposures preserve extended host emission [48]. Science-plus-sky integration totals 17.00 h; **the 67.0 h request** includes acquisition and all APT 2026.5.1 overheads, making this a Medium programme [49].

For J083826.50+371906.7 at $z = 0.2111$, ETC/Pandeia 2026.7 predicts S/N of 55/225/61 per $R = 100$ bin at rest 5/12/18 μm . These estimates assume a point source with $F_{\nu} \propto \lambda^2$, 0.95 mJy in W3 (half the archival total), the adopted sky spectrum, equal-depth sky exposures and extraction radii of 0.3''/0.6''/0.9''. They establish statistical sensitivity under the adopted nuclear-flux model; host and calibration uncertainties enter the science inference separately. Acquisition uses F560W for the faintest sources and the neutral-density filter for brighter nuclei, with readouts giving S/N above 50 without saturation over the adopted flux ranges [50].

Available optical photometry constrains the JWST-epoch state with allowance for temporal gaps; WISE/SPHEREx histories and 2010 W3/W4 photometry constrain earlier states.

Separating nuclear dust from the host. Each MRS cube measures the nucleus and surrounding host together. The PSF FWHM grows from approximately 0.3'' at 5 μm to 1'' at 25 μm , sampling roughly 0.1–4.5 kpc across the sample; channel fields span 3.2'' $\times$ 3.7'' to 6.6'' $\times$ 7.7'' [25]. Wavelength-dependent PSF fitting [51] separates unresolved and resolved emission [28]. Surrounding spectra constrain host templates with allowance for spatial variation. Joint modelling separates continuum, silicate, PAH and H₂ contributions [15,47]; PAHs allow for AGN processing [29], while [Ne V] and [Ne VI] trace highly ionised gas [46]. Alternative extractions and host templates [16] bound the nuclear flux, including compact host light within the unresolved component.

We will integrate observed-frame MRS spectra through the WISE responses, using the measured spectral shape to calculate synthetic W3/W4 fluxes in the WISE calibration convention. Comparison with 2010 requires matching the WISE spatial response: constant host light cancels in a flux difference only for the same spatial contribution. WISE imaging and aperture photometry constrain emission beyond the MRS field; optical and 2MASS profiles guide stellar-host morphology. Models account for spectral coverage gaps and host light within the SPHEREx aperture. We propagate nuclear-flux bounds from alternative extractions and host templates, uncertainty in the aperture matching, W3/W4 calibration errors and wavelength-dependent MRS calibration covariance [43].

Supplemental Information

Special requirements. Each science/sky pair is linked as a consecutive, noninterruptible sequence to obtain a contemporaneous background [48]. Sky fields are selected using WISE and 2MASS catalogues and images, with at least $25''$ clearance from catalogued sources. Twenty-three fields lie within 1–2 arcmin of their nuclei; the crowded field around J20145928+2523010 uses a $135''$ offset. Background targets are marked extended and use four-point extended-source dithers, with no target acquisition. No fixed-date constraints, coordinated parallels or joint-facility time are requested. APT’s Visit Planner returns scheduling windows for all 48 visits.

Duplications. A coordinate-based MAST search on 2026 September 26, including archived and planned JWST observations, found no overlapping observations within $120''$ of any of the 24 science positions. No potential MIRI/MRS duplications were identified.

References

- [1] Rumbaugh, N., et al. 2018, *ApJ*, 854, 160.
- [2] Graham, M. J., Ross, N. P., Stern, D., et al. 2020, *MNRAS*, 491, 4925.
- [3] Ricci, C., & Trakhtenbrot, B. 2023, *Nature Astronomy*, 7, 1282.
- [4] LaMassa, S. M., et al. 2015, *ApJ*, 800, 144.
- [5] MacLeod, C. L., et al. 2016, *MNRAS*, 457, 389.
- [6] Kishimoto, M., et al. 2013, *ApJL*, 775, L36.
- [7] Kokubo, M., & Minezaki, T. 2020, *MNRAS*, 491, 4615.
- [8] Hönig, S. F., & Kishimoto, M. 2011, *A&A*, 534, A121.
- [9] Almeyda, T., et al. 2020, *ApJ*, 891, 26.
- [10] Sheng, Z., et al. 2017, *ApJL*, 846, L7.
- [11] Stern, D., et al. 2018, *ApJ*, 864, 27.
- [12] Yang, Q., et al. 2018, *ApJ*, 862, 109.
- [13] Ross, N. P., Ford, K. E. S., Graham, M., et al. 2018, *MNRAS*, 480, 4468.
- [14] Lyu, J., & Rieke, G. 2022, *Universe*, 8, 304.
- [15] Donnan, F. R., et al. 2024, *MNRAS*, 529, 1386.
- [16] Hernán-Caballero, A., et al. 2015, *ApJ*, 803, 109.
- [17] Barvainis, R. 1987, *ApJ*, 320, 537.
- [18] Kishimoto, M., Hönig, S. F., Beckert, T., & Weigelt, G. 2007, *A&A*, 476, 713.
- [19] Koshida, S., et al. 2014, *ApJ*, 788, 159.
- [20] Landt, H., et al. 2026, *ApJ*, 997, 22.
- [21] IPAC/IRSA, “SPHEREx Data Explorer: Overview”.
- [22] Landt, H., et al. 2019, *MNRAS*, 489, 1572.
- [23] Lyu, J., & Rieke, G. H. 2021, *ApJ*, 912, 126.
- [24] Lyu, J., Rieke, G. H., & Smith, P. S. 2019, *ApJ*, 886, 33.
- [25] STScI, JWST User Documentation, “MIRI Medium Resolution Spectroscopy”.
- [26] Thompson, G. D., Levenson, N. A., Uddin, S. A., & Sirocky, M. M. 2009, *ApJ*, 697, 182.
- [27] Sirocky, M. M., et al. 2008, *ApJ*, 678, 729.
- [28] González-Martín, O., et al. 2025, *MNRAS*, 539, 2158; revised arXiv version (2026).

- [29] García-Bernete, I., et al. 2024, A&A, 691, A162.
 - [30] Panda, S., & Śniegowska, M. 2024, ApJS, 272, 13.
 - [31] Zeltyn, G., et al. 2024, ApJ, 966, 85.
 - [32] Guo, W.-J., et al. 2025, ApJS, 278, 28.
 - [33] Dong, Q., et al. 2025, ApJ, 986, 160.
 - [34] Sheng, Z., et al. 2020, ApJ, 889, 46.
 - [35] Wang, H., et al. 2026, ApJ, 997, 100.
 - [36] Hemmati, S., et al. 2026, ApJ, 998, 130.
 - [37] Laor, A., & Draine, B. T. 1993, ApJ, 402, 441; optical-property tables supplied by B. T. Draine.
 - [38] Nenkova, M., et al. 2008, ApJ, 685, 160.
 - [39] Hönig, S. F., Kishimoto, M., Tristram, K. R. W., et al. 2013, ApJ, 771, 87.
 - [40] Hönig, S. F., & Kishimoto, M. 2017, ApJL, 838, L20.
 - [41] Elitzur, M., & Shlosman, I. 2006, ApJL, 648, L101.
 - [42] Reyes-Amador, O. U., et al. 2025, MNRAS, 543, 813.
 - [43] STScI, JWST User Documentation, “MIRI MRS Calibration Status”.
 - [44] Stalevski, M., et al. 2016, MNRAS, 458, 2288.
 - [45] Mateos, S., et al. 2015, MNRAS, 449, 1422.
 - [46] Hermosa Muñoz, L., et al. 2025, A&A, 693, A321.
 - [47] Donnan, F. R., et al. 2023, MNRAS, 519, 3691.
 - [48] STScI, JWST User Documentation, “MIRI MRS Recommended Strategies” and “MIRI MRS Dedicated Sky Observations”.
 - [49] STScI, “JWST Call for Proposals for Cycle 6” and “JWST Observations Needing Special Requirements”.
 - [50] STScI, JWST User Documentation, “MIRI MRS Target Acquisition” and “JWST ETC MIRI Target Acquisition”.
 - [51] Law, D. R., et al. 2023, AJ, 166, 45.
- Generative AI disclosure.** OpenAI Codex (GPT-6) and Claude (Fable) assisted with editing the draft.

Created using the STScI L^AT_EX template for JWST Cycle 6.